

Date: 18/Feb/2026

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At $n=2$:-

$$y_3 = y_2 + h f(x_2, y_2)$$

$$y_3 = 1.32 + 0.1 [4(0.2) - 2(1.32)]$$

$$y_3 = 1.136$$

At $n=3$:-

$$y_4 = y_3 + h f(x_3, y_3)$$

$$y_4 = 1.136 + 0.1 [4(0.3) - 2(1.136)]$$

$$y_4 = 1.0288$$

At $n=4$:-

$$y_5 = y_4 + h f(x_4, y_4)$$

$$y_5 = 1.0288 + 0.1 [4(0.4) - 2(1.0288)]$$

$$y_5 = 0.98304$$

$$y(0.5) = 0.98304$$

Method 2:- Modified Euler Method

This method is a numerical technique used to approximate the solution of 1st order ODE.

$$y' = f(x, y); y(x_0) = y_0$$

Basic Euler method works on single slope. Modified Euler works on average of two slopes from beginning to end of each interval.

Formula:

$$y_{n+1}^* = y_n + h f(x_n, y_n)$$

$$y_{n+1} = y_n + \frac{h}{2} [f(x_n, y_n) + f(x_{n+1}, y_{n+1}^*)]$$

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Q- Use modified euler method to obtain a 4 decimal place approximation of indicated value. Take $h = 0.1$.

1) $y' = 2xy$; $y(1) = 1$; $y(1.5) = ?$

$f(x, y) = 2xy$; $x_0 = 1$ $y_0 = 1$ $h = 0.1$

At $n = 0$:-

$$y_1^* = y_0 + hf(x_0, y_0)$$

$$y_1^* = 1 + 0.1[2(1)(1)]$$

$$y_1^* = 1.2$$

At $n = 1$:-

$$y_2^* = y_1 + hf(x_1, y_1)$$

$$y_2^* = 1.232 + 0.1[2(1.1)(1.232)]$$

$$y_2^* = 1.5036$$

At $n = 2$:-

$$y_3^* = y_2 + hf(x_2, y_2)$$

$$y_3^* = 1.5478 + 0.1[2(1.2)(1.5478)]$$

$$y_3^* = 1.9192$$

At $n = 3$:-

$$y_4^* = y_3 + hf(x_3, y_3)$$

$$y_4^* = 1.9830 + 0.1[2(1.3)(1.9830)]$$

$$y_4^* = 2.4985$$

At $n = 4$:-

$$y_5^* = y_4 + hf(x_4, y_4)$$

$$y_5^* = 2.5905 + 0.1[2(1.4)(2.5905)]$$

$$y_5^* = 3.31592$$

Now;

$$y_1 = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^*)]$$

$$y_1 = 1 + \frac{0.1}{2} [2(1)(1) + 2(1.1)(1.2)]$$

$$y_1 = 1.232$$

Now;

$$y_2 = y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2^*)]$$

$$y_2 = 1.232 + \frac{0.1}{2} [2(1.1)(1.232) + 2(1.2)(1.5036)]$$

$$y_2 = 1.5478$$

Now;

$$y_3 = y_2 + \frac{h}{2} [f(x_2, y_2) + f(x_3, y_3^*)]$$

$$y_3 = 1.5478 + \frac{0.1}{2} [2(1.2)(1.5478) + 2(1.3)(1.9192)]$$

$$y_3 = 1.9830$$

Now;

$$y_4 = y_3 + \frac{h}{2} [f(x_3, y_3) + f(x_4, y_4^*)]$$

$$y_4 = 1.9830 + \frac{0.1}{2} [2(1.3)(1.9830) + 2(1.4)(2.4985)]$$

$$y_4 = 2.5905$$

Now;

$$y_5 = y_4 + \frac{h}{2} [f(x_4, y_4) + f(x_5, y_5^*)]$$

$$y_5 = 2.5905 + \frac{0.1}{2} [2(1.4)(2.5905) + 2(1.5)(3.31592)]$$

$$y_5 = 3.476$$

2) $y' = 1 + y^2$, $y(0) = 0$, $y(0.5) = ?$
 $f(x, y) = 1 + y^2$; $x_0 = 0$, $y_0 = 0$, $h = 0.1$

At $n = 0$:-

$$y_1^* = y_0 + h f(x_0, y_0)$$

$$y_1^* = 0 + 0.1 [1 + (0)^2]$$

$$y_1^* = 0.1$$

At $n = 1$:-

$$y_2^* = y_1 + h f(x_1, y_1)$$

$$y_2^* = 0.1 + 0.1 [1 + (0.1)^2]$$

$$y_2^* = 0.201$$

At $n = 2$:-

$$y_3^* = y_2 + h f(x_2, y_2)$$

$$y_3^* = 0.2030 + 0.1 [1 + (0.203)^2]$$

$$y_3^* = 0.307$$

At $n = 3$:-

$$y_4^* = y_3 + h f(x_3, y_3)$$

$$y_4^* = 0.3097 + 0.1 [1 + (0.3097)^2]$$

$$y_4^* = 0.4192$$

At $n = 4$:-

$$y_5^* = y_4 + h f(x_4, y_4)$$

$$y_5^* = 0.4232 + 0.1 [1 + (0.4232)^2]$$

$$y_5^* = 0.5411$$

Now:-

$$y_1 = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^*)]$$

$$y_1 = 0 + \frac{0.1}{2} [1 + (0)^2 + 1 + (0.1)^2]$$

$$y_1 = 0.1005$$

Now;

$$y_2 = y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2^*)]$$

$$y_2 = 0.1005 + \frac{0.1}{2} [1 + (0.1005)^2 + 1 + (0.201)^2]$$

$$y_2 = 0.2030$$

Now;

$$y_3 = y_2 + \frac{h}{2} [f(x_2, y_2) + f(x_3, y_3^*)]$$

$$y_3 = 0.2030 + \frac{0.1}{2} [1 + (0.2030)^2 + 1 + (0.307)^2]$$

$$y_3 = 0.3097$$

Now;

$$y_4 = y_3 + \frac{h}{2} [f(x_3, y_3) + f(x_4, y_4^*)]$$

$$y_4 = 0.3097 + \frac{0.1}{2} [1 + (0.3097)^2 + 1 + (0.4192)^2]$$

$$y_4 = 0.4232$$

Now;

$$y_5 = y_4 + \frac{h}{2} [f(x_4, y_4) + f(x_5, y_5^*)]$$

$$y_5 = 0.4232 + \frac{0.1}{2} [1 + (0.4232)^2 + 1 + (0.5411)^2]$$

$$y_5 = 0.5467$$

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4th Order Runge Kutta Method

Formula:-

$$y_{n+1} = y_n + h \left[\frac{k_1 + 2k_2 + 2k_3 + k_4}{6} \right]$$

$$k_1 = f(x_n, y_n)$$

$$k_2 = f\left(x_n + \frac{1}{2}h, y_n + \frac{1}{2}hk_1\right)$$

$$k_3 = f\left(x_n + \frac{1}{2}h, y_n + \frac{1}{2}hk_2\right)$$

$$k_4 = f(x_n + h, y_n + hk_3)$$

Q:- Use the RK-4 method with $h = 0.1$ to obtain an approximation to $y(1.3)$ for the solution of $y' = 2xy, y(1) = 1$.

Solution:-

$$f(x, y) = 2xy; \quad x_0 = 1 \quad y_0 = 1 \quad h = 0.1$$

For y at $n = 0$:-

$$k_1 = f(x_0, y_0) = 2x_0y_0 = 2(1)(1) = 2$$

$$k_2 = f\left(x_0 + \frac{h}{2}, y_0 + \frac{hk_1}{2}\right) = 2\left(x_0 + \frac{h}{2}\right)\left(y_0 + \frac{hk_1}{2}\right) = 2\left(1 + \frac{0.1}{2}\right)\left(1 + \frac{0.1 \times 2}{2}\right)$$

$$= 2.31$$

$$k_3 = f\left(x_0 + \frac{h}{2}, y_0 + \frac{hk_2}{2}\right) = 2\left(x_0 + \frac{h}{2}\right)\left(y_0 + \frac{hk_2}{2}\right) = 2\left(1 + \frac{0.1}{2}\right)\left(1 + \frac{0.1 \times 2.31}{2}\right)$$

$$= 2.34255$$

$$k_4 = f(x_0 + h, y_0 + hk_3) = 2(x_0 + h)(y_0 + hk_3) = 2(1 + 0.1)(1 + 0.1 \times 2.34255)$$

$$= 2.71536$$

Now:

$$y_1 = y_0 + \frac{h}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

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$$= \frac{1 + 0.1 [2 + 2(2.31) + 2(2.3476) + 2.7153]}{6}$$

$$y_1 = 1.233674$$

Rough table

$$x_0 = 1$$

$$y_0 = 1$$

$$x_1 = 1.1$$

$$y_1 = 1.2336$$

$$x_2 = 1.2$$

$$y_2 = 1.5526$$

For y_2 at $n=1$

$$K_1 = f(x_1, y_1) = 2(x_1, y_1) = 2(1.1)(1.2336) = 2.7141$$

$$K_2 = f\left(x_1 + \frac{h}{2}, y_1 + \frac{hK_1}{2}\right) = 2\left(x_1 + \frac{h}{2}\right)\left(y_1 + \frac{hK_1}{2}\right) = 2\left(1.1 + \frac{0.1}{2}\right)\left(1.2336 + \frac{0.1(2.71)}{2}\right)$$

$$= 3.1496$$

$$K_3 = f\left(x_1 + \frac{h}{2}, y_1 + \frac{hK_2}{2}\right) = 2\left(x_1 + \frac{h}{2}\right)\left(y_1 + \frac{hK_2}{2}\right) = 3.1496$$

$$K_4 = f(x_1 + h, y_1 + hK_3) = 2(x_1 + h)(y_1 + hK_3) = 3.7288$$

$$y_2 = y_1 + \frac{h}{6} (K_1 + 2K_2 + 2K_3 + K_4)$$

$$y_2 = 1.5526$$

$$x_3 = 1.3$$

$$y_3 = 1.9937$$

For y_3 at $n=2$

$$K_1 = f(x_2, y_2) = 2(x_2, y_2) = 3.7264$$

$$K_2 = f\left(x_2 + \frac{h}{2}, y_2 + \frac{hK_1}{2}\right) = 2\left(x_2 + \frac{h}{2}\right)\left(y_2 + \frac{hK_1}{2}\right) = 4.3475$$

$$K_3 = f\left(x_2 + \frac{h}{2}, y_2 + \frac{hK_2}{2}\right) = 2\left(x_2 + \frac{h}{2}\right)\left(y_2 + \frac{hK_2}{2}\right) = 4.424$$

$$K_4 = f(x_2 + h, y_2 + hK_3) = 2(x_2 + h)(y_2 + hK_3) = 5.1872$$

$$y_3 = y_2 + \frac{h}{6} (K_1 + 2K_2 + 2K_3 + K_4)$$

$$y_3 = 1.9937$$

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Q:- Use RK4 method with $h = 0.1$ to obtain a four decimal approximation of the indicated value.

$$y' = 2x - 3y + 1; y(1) = 5; y(1.3) = ?$$

Solution:-

For y_1 at $n = 0$:-

$$k_1 = f(x_0, y_0) = 2(x_0) - 3(y_0) + 1 = -12$$

$$k_2 = f\left(x_0 + \frac{h}{2}, y_0 + \frac{hk_1}{2}\right) = 2\left(x_0 + \frac{h}{2}\right) - 3\left(y_0 + \frac{hk_1}{2}\right) + 1 = -10.1$$

$$k_3 = f\left(x_0 + h, y_0 + hk_1\right) = 2\left(x_0 + h\right) - 3\left(y_0 + hk_1\right) + 1 = -16.355$$

$$k_4 = f\left(x_0 + h, y_0 + hk_3\right) = 2\left(x_0 + h\right) - 3\left(y_0 + hk_3\right) + 1 = -8.6845$$

$$y_1 = y_0 + \frac{h}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$y_1 = 3.9734$$

For y_2 at $n = 1$:-

$$k_1 = 2(x_1) - 3(y_1) + 1 = -8.7202$$

$$k_2 = 2\left(x_1 + \frac{h}{2}\right) - 3\left(y_1 + \frac{hk_1}{2}\right) + 1 = -7.3121$$

$$k_3 = 2\left(x_1 + h\right) - 3\left(y_1 + hk_1\right) + 1 = -7.5233$$

$$k_4 = 2(x_1 + h) - 3(y_1 + hk_3) + 1 = -6.2631$$

$$y_2 = y_1 + \frac{h}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$y_2 = 3.2291$$

$$\begin{array}{ll} x_0 = 1 & y_0 = 5 \\ x_1 = 1.1 & y_1 = 3.9734 \\ x_2 = 1.2 & y_2 = 3.2291 \end{array}$$

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$$x_3 = 1.3 \quad y_3 = 2.6$$

for y_3 at $n=2$:-

$$k_1 = 2(x_2) - 3(y_2) + 1 = -6.2852$$

$$k_2 = 2\left(x_2 + \frac{h}{2}\right) - 3\left(y_2 + \frac{hk_1}{2}\right) + 1 = -5.24242$$

$$k_3 = 2\left(x_2 + \frac{h}{2}\right) - 3\left(y_2 + \frac{hk_2}{2}\right) + 1 = -5.3988$$

$$k_4 = 2(x_2 + h) - 3(y_2 + hk_3) + 1 = -4.4655$$

$$y_3 = 3.2284 + \frac{h}{6} [k_1 + 2k_2 + 2k_3 + k_4]$$

$$y_3 = 2.6911$$

Laplace Equation:-

$$U_{xx} + U_{yy} = 0, \quad \frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} = 0$$

After sol of PDE by NM

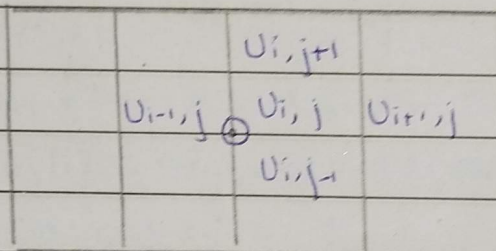
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Determine Steady State Temp distribution in thin square plate.

Consider a square region with the values of 'U' temperature given at the grid points on the boundary. We want to solve Laplace equation with given boundary conditions and obtain the values of 'U' at the interior grid points.

If we consider a square mesh i.e. h-k, then we have



* avg n kaha hota above, below, right, left ke.

Standard 5-point formula:-

$$U_{i,j} = \frac{1}{4} (U_{i+1,j} + U_{i-1,j} + U_{i,j+1} + U_{i,j-1})$$

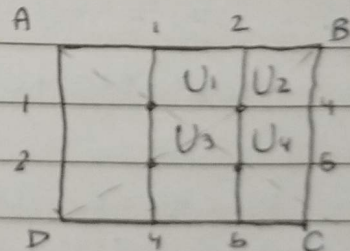
Q:- Solve $U_{xx} + U_{yy} = 0$ correct upto 2 decimal places at nodal points of following square grid using boundary value indicated.

Sol:- $U_2 = U_3$

Eq for U_1 :- $U_1 = \frac{1}{4} (1 + 1 + U_2 + U_3) = \frac{1}{4} (2 + U_2 + U_3)$

Eq for U_2 :- $U_2 = \frac{1}{4} (2 + 4 + U_1 + U_4) = \frac{1}{4} (6 + U_1 + U_4)$

Eq for U_4 :- $U_4 = \frac{1}{4} (5 + 5 + U_2 + U_3) = \frac{1}{4} (10 + 2U_2)$



Apply Gauss Seidel Iteration method;

$$U_1 = 1.999$$

$$U_3 = 3.999$$

$$U_2 = U_4 = 2.999$$

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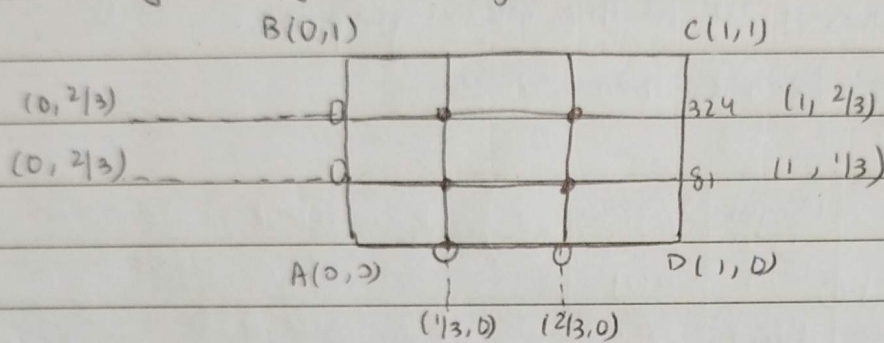
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Q:- Solve $U_{xx} + U_{yy} = 0$ with $h = k = \frac{1}{3}$ in a square mesh of 9 grids with vertices $A(0,0)$, $B(0,1)$, $C(1,1)$ and $D(1,0)$ given that the unknown function on boundary is $U(x,y) = 729x^2y^2$.



Sol:-

$$U_1 = U_4$$

$$U_1 = \frac{1}{4} (0 + 81 + U_2 + U_3)$$

$$U_4 = U_1 = \frac{1}{4} (81 + U_2 + U_3)$$

$$U_2 = \frac{1}{4} (324 + 324 + U_1 + U_4) = \frac{1}{4} (648 + 2U_1)$$

$$U_3 = \frac{1}{4} (0 + 0 + U_1 + U_4)$$

$$U_3 = \frac{1}{4} (2U_1)$$

Apply Gauss Seidel,

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Solution of PDE by Numerical Method:-

Finite difference techniques are used for the solution of 2 dimensional Laplace and Poisson equations on a rectangular domain.

Solution of 1 dimensional heat equations can be obtained by using Bender Schniedt and Grant Nicholson difference schemes (methods)

Classification of PDE's:-

General form of PDE:-

$$AU_{xx} + BU_{xy} + CU_{yy} + DU_x + EU_y + FU = 0$$

where A, B, C, D, E, F are the function of x and y.

$\Rightarrow$ Eq $U^n(1)$ is elliptic in a plane if $B^2 - 4AC < 0$.

$\Rightarrow$ Eq $U^n(1)$ is parabolic in a plane if $B^2 - 4AC = 0$.

$\Rightarrow$ Eq $U^n(1)$ is hyperbolic in a plane if $B^2 - 4AC > 0$.

Classify the following PDE's:-

1) $U_{xx} - 2xy + U_{yy} = 0$

$$B^2 - 4AC = 0$$

$$4 - 4(1)(1) = 0$$

$$4 - 4 = 0$$

$$0 = 0 \text{ PARABOLIC}$$

2) $U_{xx} - 4U_{xy} = 0$

$$B^2 - 4AC = 0 ; 16 - 4(1)(1) = 0$$

$$16 - 4 = 0 \quad 12 > 0 \text{ HYPERBOLIC}$$

3) $U_{xx} + U_{yy} = 0$ (Laplace eq)

$$B^2 - 4AC = 0 ; 0^2 - 4(1)(1) = 0$$

$$0 - 4 < 0$$

$$-4 < 0$$

Elliptic

Other famous PDE's:-

1) 1-D heat eq: $U_{xx} = a^2 U_y$ Parabolic

2) 1-D wave eq: $U_{xx} = a^2 U_{yy}$ Hyperbolic

1) $U_{xx} - a^2 U_y = 0$

$$B^2 - 4AC = 0 ; 0^2 - 4(A)(0) = 0$$

$$0 = 0 \text{ PARABOLIC}$$

2) $U_{xx} = a^2 U_{yy} = 0$

$$B^2 - 4AC = 0$$

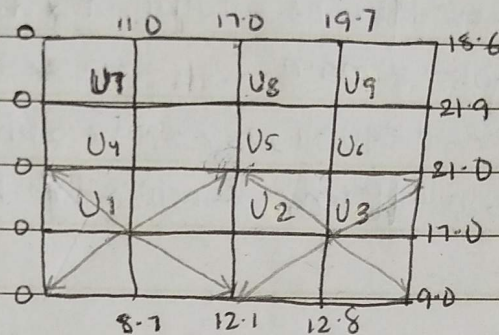
$$0 - 4(A)(-a^2) = 0$$

$$4a^2 > 0 \text{ HYPERBOLIC}$$

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Q:- Solve the Laplace eqn $\nabla^2 U = 0$ at the interior points of the least square given in fig.



SOLUTION:-

Initially we will find rough values of interior points by and then we will improve those values by applying SPFF with Gauss Seidel iteration.

Rough values:- bakival unknown hai is like a box chors etc.

$$U_5 = \frac{1}{4} (11.0 + 12.1 + 21.0) = 12.525 \quad (\text{SPFF})$$

* between SPFF

we will find U_1, U_3, U_7 & U_9 by applying DFPP.

* 4 corner DFPP

$$U_1 = \frac{1}{4} (10.0 + 0 + 12.1 + 12.525) = 6.15$$

* back gave SPFF

$$U_3 = 13.65 ; U_7 = 7.38 ; U_9 = 17.53$$

* rough values nikalne

we will find U_2, U_4, U_6, U_8 by SPFF

ke baad jithe bli

$$U_2 = \frac{1}{4} (12.1 + 0 + U_3 + U_1) = 11.10$$

interior points wpe SPFF lagare hai

Next

$$U_4 = 6.51$$

$$U_6 = 16.17$$

$$U_8 = 13.60$$

Now; apply SPFF interior points:- Improved values-

$$U_1 = \frac{1}{4} (U_4 + U_2 + 8.7 + 0) = 6.577 ; U_9 = 17.829$$

$$U_2 = 11.21 ; U_3 = 14.279 ; U_4 = 6.618 ; U_5 = 11.864 ; U_6 = 16.115 ; U_7 = 7.829 ; U_8 = 13.523$$

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Applying Gauss Seidel

Iter no	U_1	U_2	U_3	U_4	U_5	U_6	U_7	U_8	U_9	
0	6.15	11.10	13.65	6.51	12.52	16.17	7.38	13.60	17.53	
1	6.67	11.21	14.27	6.61	11.88	16.115	7.829	13.503	17.821	
2	6.63	11.22	14.28	6.58	11.85	16.236	7.797	13.614	17.862	
3	6.62	11.21	14.31	6.56	11.90	16.27	7.820	13.647	17.879	

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CHP 5:- Difference Operator

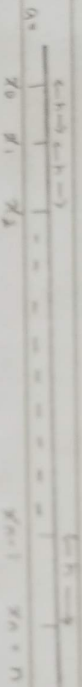
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Introduction:-

Let us consider a function $y = f(x)$, defined on interval (a, b) . Let us consider discrete value of x differing by equal length 'h', i.e. $a = x_0, x_1, x_2, \dots, x_{n-1}, x_n = b$ or

$$x_k = x_0 + kh \quad h = 0, 1, 2, 3, \dots, n$$

It is used to study changes in values specifically when data are discrete.



Let the corresponding value of the function are,

$$y_0 = f_0 = y(x_0) = f(x_0)$$

$$y_1 = f_1 = y(x_0 + h) = f(x_0 + h)$$

$$y_2 = f_2 = y(x_0 + 2h) = f(x_0 + 2h)$$

$$\vdots$$
$$y_n = f_n = y(x_0 + nh) = f(x_0 + nh)$$

The values of x are called the arguments and the values of the function $y = f(x)$, corresponding to the arguments are called entries.

LINEAR OPERATOR:-

An operator 'L' is linear, for any function $f(x)$ and $g(x)$ and constants α and β , it contains

two properties:

1) Additivity: $L[f(x) + g(x)] = L[f(x)] + L[g(x)]$

2) Homogeneity: $L[\alpha f(x)] = \alpha L[f(x)]$

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Examples of linear Operator:

1) Differentiation: $D = d/dx$

2) Integration: $I[f(x)] = \int f(x) dx$

3) Finite Difference Operator:

$$f \rightarrow \Delta$$

$$B \rightarrow \nabla$$

$$C \rightarrow \delta$$

4) Shift Operator: $L[f(x)] = f(x+h)$

5) Identity operator

Forward Diff Operator (Δ):-

The first order operator is denoted by (Δ) and defined by:

$$\Delta y_0 = y_1 - y_0$$

$$\Delta y_1 = y_2 - y_1$$

$$\Delta y_2 = y_3 - y_2$$

!

$$\Delta y_{n-1} = y_n - y_{n-1}$$

The second order difference operator is defined by:

$$\Delta^2 y_0 = \Delta y_1 - \Delta y_0$$

$$\Delta^2 y_1 = \Delta y_2 - \Delta y_1$$

$$\Delta^2 y_2 = \Delta y_3 - \Delta y_2$$

!

$$\Delta^2 y_{n-2} = \Delta y_{n-1} - \Delta y_{n-2}$$

The third order forward difference operator is defined by,

$$\Delta^3 y_0 = \Delta^2 y_1 - \Delta^2 y_0$$

$$\Delta^3 y_1 = \Delta^2 y_2 - \Delta^2 y_1$$

$$\Delta^3 y_2 = \Delta^2 y_3 - \Delta^2 y_2$$

⋮

$$\Delta^3 y_{n-3} = \Delta^2 y_{n-2} - \Delta^2 y_{n-3}$$

Forward Difference Table:-

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
x_0	y_0	Δy_0			
x_1	y_1	$\rightarrow \Delta y_1$	$\rightarrow \Delta^2 y_0$		
x_2	y_2	$\rightarrow \Delta y_2$	$\rightarrow \Delta^2 y_1$	$\rightarrow \Delta^3 y_0$	
x_3	y_3	$\rightarrow \Delta y_3$	$\rightarrow \Delta^2 y_2$	$\rightarrow \Delta^3 y_1$	$\rightarrow \Delta^4 y_0$
x_4	y_4				

* In forward difference table upper diagonal is conserved.

Q:- Construct the forward difference table for the following data:

x	0	1	2	3	4	5
y	12	15	20	27	39	52

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x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
0	12					
		→ 3				
1	15		→ 2			
		→ 5		→ 0		
2	20		→ 2		→ 3	
		→ 7		→ 3		→ -10
3	27		→ 5		→ -7	
		→ 12		→ -4		
4	39		→ 1			
		→ 13				
5	52					

Q - Given $y = x^2 + 5x + 7$, evaluate for $x = 0, 1, 2, 3, 4$ then construct forward difference table

x	0	1	2	3	4
y	7	13	21	31	43

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
0	7			
		6		
1	13		2	
		8		0
2	21		2	
		10		0
3	31		2	
		12		
4	43			

Q. Express $\Delta^2 y_0$ and $\Delta^3 y_0$ in terms of the value of function y .

$$\begin{aligned}\Delta^2 y_0 &= \Delta y_1 - \Delta y_0 \\ &= (y_2 - y_1) - (y_1 - y_0) \\ &= y_2 - y_1 - y_1 + y_0 \\ \Delta^2 y_0 &= y_2 - 2y_1 + y_0\end{aligned}$$

$$\begin{aligned}\Delta^3 y_0 &= \Delta^2 y_1 - \Delta^2 y_0 \\ &= (\Delta y_2 - \Delta y_1) - (\Delta y_1 - \Delta y_0) \\ &= [(y_3 - y_2) - (y_2 - y_1)] - [(y_1 - y_0) - (y_0 - y_{-1})] \\ &= y_3 - y_2 - y_2 + y_1 - (y_1 - y_0 + y_0 - y_{-1}) \\ &= y_3 - 2y_2 + y_1 - (y_1 - y_0 + y_0 - y_{-1}) \\ &= y_3 - 2y_2 + y_1 - y_1 + y_0 + y_{-1} - y_0 + y_{-1} \\ \Delta^3 y_0 &= y_3 - 3y_2 + 3y_1 - y_0\end{aligned}$$

Backward Difference Table -

The first order backward difference operator is denoted by ' ∇ ' and defined by:

$$\begin{aligned}\nabla y_1 &= y_1 - y_0 \quad \text{curr - prev} \\ \nabla y_2 &= y_2 - y_1 \\ \nabla y_3 &= y_3 - y_2 \\ &\vdots \\ \nabla y_n &= y_n - y_{n-1}\end{aligned}$$

For second order,

$$\begin{aligned}\nabla^2 y_2 &= \nabla y_2 - \nabla y_1 \\ \nabla^2 y_3 &= \nabla y_3 - \nabla y_2 \\ &\vdots \\ \nabla^2 y_n &= \nabla y_n - \nabla y_{n-1}\end{aligned}$$

For third order,

$$\begin{aligned}\nabla^3 y_3 &= \nabla^2 y_3 - \nabla^2 y_2 \\ \nabla^3 y_4 &= \nabla^2 y_4 - \nabla^2 y_3 \\ \nabla^3 y_n &= \nabla^2 y_n - \nabla^2 y_{n-1}\end{aligned}$$

Backward Difference Table

x_0	y_0	∇y_0	$\nabla^2 y_0$	$\nabla^3 y_0$	$\nabla^4 y_0$
x_1	y_1	∇y_1	$\nabla^2 y_1$	$\nabla^3 y_1$	
x_2	y_2	∇y_2	$\nabla^2 y_2$	$\nabla^3 y_2$	$\nabla^4 y_2$
x_3	y_3	∇y_3	$\nabla^2 y_3$	$\nabla^3 y_3$	
x_4	y_4	∇y_4			

* lower diagonal
conserved

Q:- Construct backward difference table for following data

x	0	1	2	3	4
y	2	3	12	35	78

x	0	1	2	3	4
y	2	3	12	35	78
∇y		1	9	23	43
$\nabla^2 y$			8	14	20
$\nabla^3 y$				6	6
$\nabla^4 y$					0

Q:- Construct Backward difference table

z	10	20	30	40	SD
y	1	1.3010	1.4771	1.6021	1.6990

$$z \quad y \quad \nabla y \quad \nabla^2 y \quad \nabla^3 y \quad \nabla^4 y$$

$$10 \quad 1$$

$$20 \quad 1.3010$$

$$0.1761$$

$$-0.1244$$

$$0.07$$

$$30 \quad 1.4771$$

$$-0.0511$$

$$0.125$$

$$0.023$$

$$-0.0506$$

$$40 \quad 1.6021$$

$$-0.0781$$

$$0.0964$$

$$SD \quad 1.6990$$

Central Difference Operator:-

The first order central difference operator is denoted by δ and defined by,

$$\delta y_{1/2} = y_1 - y_0$$

$$\delta y_{3/2} = y_2 - y_1$$

$$\delta y_{5/2} = y_3 - y_2$$

⋮

$$\delta y_{n-1/2} = y_n - y_{n-1}$$

Similarly for high order,

$$\delta^{(n)} y_{k+1/2} = \delta^{(n-1)} y_{k+1} - \delta^{(n-1)} y_k$$

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Central Difference Table:

x_0	y_0	δy	$\delta^2 y$	$\delta^3 y$	$\delta^4 y$
x_1	y_1	$-\delta y^{1/2}$	$-\delta y^1$	$-\delta y^{3/2}$	$-\delta y^2$
x_2	y_2	$-\delta y^{3/2}$	$-\delta y^2$	$-\delta y^{5/2}$	$-\delta y^3$
x_3	y_3	$-\delta y^{5/2}$	$-\delta y^3$		
x_4	y_4				

Q1- Construct a central difference table for:-

a)

x	$y=f(x)$	δ	δ^2	δ^3	δ^4	δ^5	δ^6
0	0.000	0	0.191	0.341	0.433	0.477	0.499

x	y	δy	$\delta^2 y$	$\delta^3 y$	$\delta^4 y$	$\delta^5 y$	$\delta^6 y$
0.000	0	0.191					
0.5	0.191		-0.041				
1.0	0.341	0.15	-0.058				
1.5	0.433	0.092	-0.048	0.01			
2	0.477	0.044	0.02	0.01	-0.005		
2.5	0.499	0.017	-0.0027	0.0015			
3	0.499		-0.002				

b)

x	0	1	2	3	4	5	6
y	0	1	26	63	124	342	511

x y $s_0 y$ $s^2 y$ $s^3 y$ $s^4 y$ $s^5 y$ $s^6 y$

1	7	12					
	19		6				
2	26	18		0			
	37		6		127		
3	63	24		127		-593	
	61		133		-466		
4	124	157		-334			
	218		-200				
5	342	-49					
	164						
6	511						

Q: Prove that $\Delta^3 y_2 = \nabla^3 y_5$

$$\Delta^3 y_2 = \nabla^3 y_5$$

$$\Delta^2 y_3 - \Delta^2 y_2 = \nabla^2 y_5 - \nabla^2 y_4$$

$$\Delta y_4 - \Delta y_3 - \Delta y_3 + \Delta y_2 = \nabla y_5 - \nabla y_4 - \nabla y_4 + \nabla y_3$$

$$y_5 - y_4 - (y_4 - y_3) - (y_4 - y_3) + y_3 - y_2 = y_5 - y_4 - (y_4 - y_3) - (y_4 - y_3) + y_3 - y_2$$

$$y_5 - y_4 - y_4 + y_3 - y_4 + y_3 + y_3 - y_2 = y_5 - y_4 - y_4 + y_3 - y_4 + y_3 + y_3 - y_2$$

$$y_5 - 3y_4 + 3y_3 - y_2 = y_5 - 3y_4 + 3y_3 - y_2$$

$$0 = 0$$

Hence proved!

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Shift Operator:-

- The shift operator 'E' is defined as;

$$Ef(x) = f(x+h) \text{ or } Ey(x) = y(x+h)$$

Similarly higher order shift operator is defined by;

$$E^2 f(x) = f(x+2h)$$

$$E^3 f(x) = f(x+3h)$$

$$\vdots$$

$$E^n f(x) = f(x+nh)$$

We have;

$$Ey_0 = Ey(x_0) = y(x_0+h) = y_1$$

So,

$$E^2 y_0 = y_2$$

$$E^3 y_0 = y_3$$

$$\vdots$$

$$E^n y = y_n$$

Inverse Shift Operator: It is defined as;

$$E^{-1} f(x) = f(x-h)$$

$$\text{or } E^{-n} f(x) = f(x-nh)$$

Relation between E and Δ :-

By definition of forward difference operator;

$$\Delta f(x) = f(x+h) - f(x)$$

$$\Delta y_0 = y_1 - y_0$$

$$\Delta f(x) = E f(x) - f(x)$$

$$\Delta = E - I$$

$$E = \Delta + I$$

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Relationship between E and ∇ :-

By definition of backward difference operator,

$$\nabla f(x) = f(x) - f(x-h)$$

$$\nabla f(x) = f(x) - E^{-1}f(x)$$

$$\nabla = 1 - E^{-1}$$

OR

$$E^{-1} = 1 - \nabla$$

$$E = (1 - \nabla)^{-1}$$

Relationship between E and S :-

By definition of central difference operator,

$$Sf(x) = f\left(x + \frac{h}{2}\right) - f\left(x - \frac{h}{2}\right)$$

$$Sf(x) = E^{1/2} \cdot f(x) - E^{-1/2}f(x)$$

$$S = E^{1/2} - E^{-1/2}$$

Differential Operator:-

$$D = \frac{d}{dx} f(x) = f'(x)$$

$$D^2 = \frac{d^2}{dx^2} f(x) = f''(x)$$

⋮

$$D^n = \frac{d^n}{dx^n} f(x) = f^{(n)}(x)$$

Unit Operator:-

$$1f(x) = f(x)$$

Zero Operator:-

$$0f(x) = 0$$

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CHP 6: INTERPOLATION & CURVE FITTING

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INTERPOLATION = Interpolation is a process of finding the value of function for any value of independent variable in an interval (x_0, x_n) .

POLYNOMIAL INTERPOLATION = Approximation by the polynomial of least degree satisfying the given data is called polynomial interpolation and the corresponding polynomial is called interpolating polynomial.

1) NEWTON

FORWARD DIFFERENCE INTERPOLATION =

- Formula:-

$$y(x) = p(x) = y_0 + P \Delta y_0 + \frac{P(P-1) \Delta^2 y_0}{2!} + \frac{P(P-1)(P-2) \Delta^3 y_0}{3!} + \dots + \frac{P(P-1)(P-2) \dots (P-n+1) \Delta^n y_0}{n!}$$

where;

$$P = \frac{x - x_0}{h}$$

Q:- Find the forward difference polynomial for data.

x	-1	0	1	2
y	4	2	2	4

$h = 1$, step size

Solution;

$$P = \frac{x - x_0}{h} = \frac{x - (-1)}{1} = x + 1$$

Here;

$$y_0 = 4; \Delta y_0 = 2 - 4 = -2; \Delta^2 y_0 = 2; \Delta^3 y_0 = 0$$

By applying formula;

$$y(x) = p(x) = x^2 - x + 4$$

Estimate $y(-0.5)$

$$y(-0.5) = (-0.5)^2 - (-0.5) + 4 \Rightarrow 2.75$$

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a) NEWTON BACKWARD INTERPOLATION POLYNOMIAL FORMULA:-

Consider,

$$P(x) = f(x) = y(x_0 + ph) = E^p y(x_0) = E^p y_n \quad \because x = x_0 + ph$$

$$P(x) = (1 - \nabla)^p y_n \quad \because E = (1 - \nabla)^{-1}$$

Again using binomial formula for negative power, we have

$$y(x) = P(x) = y_n + p \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \dots + \frac{p(p+1)(p+2) \dots (p+n-1)}{n!} \nabla^n y_n$$

where $p = \frac{x - x_0}{h}$

Q - Estimate $y(42)$ from the following data using Newton's backward interpolation formula.

x	20	25	30	35	40	45
y	354	332	291	265	231	204

Backward Difference Table:-

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$	$\nabla^5 y$
20	354					
		-22				
25	332		-19			
		-41	29			
30	291		10	-31		
		-31	-8	45		
35	265		2	0	8	
		-29	2	0		
40	231		-21	2		

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From the table above, we have

$$y_n = 204, \nabla y_n = -27, \nabla^2 y_n = 2, \nabla^3 y_n = 6, \nabla^4 y_n = 8, \nabla^5 y_n = 45$$

Here $x_n = 45$, $x = 42$ and $h = 5$ then

$$\Rightarrow p = \frac{x - x_n}{h} = \frac{42 - 45}{5} = \frac{-3}{5} = -0.6$$

As the point $x = 42$ is near the end of the table we use Newton's backward interpolation formula;

$$P(x) = y_n + p \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_n +$$

$$\frac{p(p+1)(p+2)(p+3)}{4!} \nabla^4 y_n + \frac{p(p+1)(p+2)(p+3)(p+4)}{5!} \nabla^5 y_n$$

Substituting values, we get.

$$\Rightarrow P(42) = 204 + (-0.6)(-27) + \frac{(-0.6)(10.4)}{2} (2) + \frac{(-0.6)(10.4)(11.4)}{6} (6) +$$

$$\frac{(-0.6)(10.4)(11.4)(12.4)}{24} (8) + \frac{(-0.6)(10.4)(11.4)(12.4)(13.4)}{120} (45)$$

$$y(42) = f(42) \approx P(42) = 218.6630$$

Practice
Q5

← +1.5 ke liye forward coz near to initial val
and 1.7 ke liye backward coz near to last val.

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Lagrange's Interpolation Polynomial Formula:-

The Lagrange's interpolation polynomial formula of degree at most

$(n+1)$ is given by:

$$y(x) \approx f(x) \approx P(x) = \sum_{i=0}^n L_i(x) y_i = L_0(x) y_0 + L_1(x) y_1 + \dots + L_n(x) y_n$$

where $L_i(x)$ are called Lagrange's interpolation coefficients and defined as:

$$L_i(x) = \frac{(x-x_0)(x-x_1)\dots(x-x_{i-1})(x-x_{i+1})\dots(x-x_n)}{(x_1-x_0)(x_1-x_1)\dots(x_1-x_{i-1})(x_1-x_{i+1})\dots(x_1-x_n)}$$

$$(x_1-x_0)(x_1-x_1)\dots(x_1-x_{i-1})(x_1-x_{i+1})\dots(x_1-x_n)$$

Example:- Determine the Lagrange's interpolation polynomial

for the data:

x	x_0	x_1	x_2
	0	1	3
$y=f(x)$	1	3	65

and assume the value of $y=f(x)$ at $x=2$ and $x=4$

Solution:-

The Lagrange's interpolation polynomial formula of degree at most 2 is

given by (here $n=2$):-

$$y(x) \approx f(x) \approx P(x) = \sum_{i=0}^2 L_i(x) y_i$$

$$\Rightarrow y(x) \approx f(x) \approx P(x) = L_0(x) y_0 + L_1(x) y_1 + L_2(x) y_2 \quad \text{--- (1)}$$

Now,

$$L_0(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} = \frac{(x-1)(x-3)}{(0-1)(0-3)} = \frac{1}{3} (x^2 - 4x + 3)$$

$$(x_0-x_1)(x_0-x_2) = (0-1)(0-3)$$

$$L_1(x) = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} = \frac{(x-0)(x-3)}{(1-0)(1-3)} = -1 (x^2 - 3x)$$

$$(x_1-x_0)(x_1-x_2) = (1-0)(1-3)$$

$$L_2(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} = \frac{(x-0)(x-1)}{(3-0)(3-1)} = \frac{1}{6} (x^2 - x)$$

$$(x_2-x_0)(x_2-x_1) = (3-0)(3-1)$$

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Substituting values in eq (1), we get

$$y(x) = f(x) \cong \frac{P(x) = 1}{3} (x^2 - 4x + 3)(1) - \frac{1}{2} (x^2 - 3x)(3) + \frac{1}{6} (x^2 - x)(155)$$

$$y = f(x) = P(x) = 8x^2 - 6x + 1$$

is the required Lagrange's interpolation polynomial. $A(15)$

$$y(12) = f(12) \cong P(12) = 8(12)^2 - 6(12) + 1 = 21$$

and

$$y(4) = f(4) \cong P(4) = 8(4)^2 - 6(4) + 1 = 105$$

Newton Divided Difference Interpolation

$$y(x) = f(x) = f(x_0) + (x - x_0) f[x_0, x_1] + (x - x_0)(x - x_1) f[x_0, x_1, x_2] + (x - x_0)(x - x_1)(x - x_2) f[x_0, x_1, x_2, x_3] + \dots + (x - x_{n-1})$$

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CUBIC SPLINE INTERPOLATION-

Formulas:-

$$\Rightarrow S(x) = \frac{1}{6h} [(x_i - x)^3 m_{i-1} + (x - x_i)^3 m_i] + \frac{1}{h} (x_i - x) \left[y_{i-1} - \frac{h^2}{6} m_{i-1} \right] + \frac{1}{h} (x - x_{i-1}) \left[y_i - \frac{h^2}{6} m_i \right]$$

$$\Rightarrow m_{i-1} + 4m_i + m_{i+1} = \frac{6}{h^2} [y_{i-1} + 2y_i + y_{i+1}] \rightarrow \text{for } i = (n-1) \text{ times}$$

$$\Rightarrow m_0 = m_n = 0$$

Q:- From the following table:

x	1	2	3
y	-8	-1	18

Compute $y(1.5)$ and $y'(1)$ using cubic spline?

Sol:-

$$h=1 \quad n=2 \quad m_0 = 0 \quad m_2 = 0$$

$i=1:-$

$$m_0 + 4m_1 + m_2 = \frac{6}{1} [y_0 - 2y_1 + y_2]$$

$$0 + 4m_1 + 0 = 6[-8 - 2(-1) + 18]$$

$$m_1 = 18$$

for $i=1:-$

$$S(x) = \frac{1}{6} [(x_1 - x)^3 m_0 + (x - x_0)^3 m_1] + \frac{1}{1} (x_1 - x) \left[y_0 - \frac{1}{6} m_0 \right] + \frac{1}{1} (x - x_0) \left[y_1 - \frac{1}{6} m_1 \right]$$

$$S(x) = \frac{1}{6} [0 + (x-1)^3 (18) + 1(2-x)(-8-0) + \frac{1}{1} (x-1) \left[-1 - \frac{1}{6} (18) \right]]$$

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$$S(x) = \frac{1}{6} [18(x-1)^3] + (2-x)(1-8) + (x-1)(1-4)$$

$$S(x) = 3x^3 - 9x^2 + 13x - 15$$

$$S(1.5) = -5.6$$

$$S'(x) = 9x^2 - 18x + 13$$

$$S'(1) = 4$$

Q:- Find the Cubic Spline interpolation polynomial for the data given below under the condition $M(0) = 0 = M(3)$ and valid in the sub-interval $[1, 2]$. Hence find $f(1.5)$.

x	0	1	2	3
$y = f(x)$	1	4	10	8

Solution:-

$$h = 1; M_0 = 0; M_3 = 0$$

$i = 1$:-

$$M_{i-1} + 4M_i + M_{i+1} = \frac{6}{h^2} [y_{i-1} - 2y_i + y_{i+1}]$$

$$M_0 + 4M_1 + M_2 = \frac{6}{1^2} [y_0 - 2y_1 + y_2]$$

$$0 + 4M_1 + M_2 = \frac{6}{1} [1 - 2(4) + 10]$$

$$4M_1 + M_2 = 6(3)$$

$$4M_1 + M_2 = 18 \quad \text{--- (1)}$$

$i = 2$:-

$$M_1 + 4M_2 + M_3 = \frac{6}{h^2} [y_1 - 2y_2 + y_3]$$

$$M_1 + 4M_2 + D = \frac{6}{12} [4 - 2(10) + 8]$$

$$M_1 + 4M_2 = 6(-8)$$

$$M_1 + 4M_2 = -48 \quad \text{--- (2)}$$

Solving eq (1) and eq (2),

$$4M_1 + M_2 = 18 \quad \text{--- (1)}$$

$$M_1 + 4M_2 = -48 \quad \text{--- (2)}$$

$$\textcircled{1} \Rightarrow M_2 = 18 - 4M_1$$

Substitute in eq (2),

$$\text{eq (2)} \Rightarrow M_1 + 4(18 - 4M_1) = -48$$

$$M_1 + 72 - 16M_1 = -48$$

$$-15M_1 = -120$$

$$\boxed{M_1 = 8}$$

$$\text{eq (1)} \Rightarrow 4(8) + M_2 = 18$$

$$M_2 + 32 = 18$$

$$\boxed{M_2 = -14}$$

Now, for spline on interval [1, 2] :-

$\rightarrow i=2$ (upper bound)

$$S(x) = \frac{1}{6} [(x_1 - x)^3 M_{i-1} + (x - x_{i-1})^3 M_i] + \frac{1}{h} (x_1 - x) \left[y_{i-1} - \frac{h^2}{6} M_{i-1} \right] + \frac{1}{h} (x - x_{i-1}) \left[y_i - \frac{h^2}{6} M_i \right]$$

Ch

$$= \frac{1}{6} [(x_2 - x)^3 M_1 + (x - x_1)^3 M_2] + \frac{1}{h} (x_2 - x) \left[y_1 - \frac{h^2}{6} M_1 \right] + \frac{1}{h} (x - x_1) \left[y_2 - \frac{h^2}{6} M_2 \right]$$

$$= \frac{1}{6} [(2-x)^3 (8) + (x-1)^3 (-14)] + \frac{1}{1} (2-x) \left[4 - \frac{1}{6} (8) \right] + \frac{1}{1} (x-1) \left[10 - \frac{1}{6} (-14) \right]$$

$$S(x) = \frac{1}{6} [8(2-x)^3 - 14(x-1)^3] + (2-x) \frac{8}{3} + (x-1) \frac{31}{3} \quad \text{--- (3)}$$

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At $f(1.5)$:-

$$x = 1.5$$

$$\rightarrow (2-x) = (2-1.5) = 0.5$$

$$\rightarrow (x-1) = (1.5-1) = 0.5$$

$$eqn (A) \rightarrow \frac{1}{6} [8(1.5-1)^3 - 14(1.5-1)^2 + 10(1.5-1) + \frac{(3.1)}{3}(1.5-1)]$$

$$= \frac{1}{6} [8(1-1) + 14(1-1) + 10(0.5) + \frac{3.1}{3}(0.5)]$$

$$= \frac{1}{6} [0 + 0 + 5 + \frac{1.55}{3}]$$

$$= 0.16667 - 0.2917 + 1.333 + 0.1667$$

$$f(1.5) = 1.375$$

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Curve fitting on a straight line: $y = ax + b$

$$y = ax + b$$

$$\sum y = a \sum x + nb \quad \text{--- (i)}$$

Multiply by 'x' on both sides:

$$\sum xy = a \sum x^2 + b \sum x \quad \text{--- (ii)}$$

Q. By the method of least square find the straight line that best fits the following

data:-

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dekhana hoga..

x	1	2	3	4	5
y	14	27	40	55	68

Calc
- click on mode then press 3 (STAT)

choose 2 (A+BX). Put val of x and y

then press =. After that press AC

- press shift 1 → choose 5 (reg)

now choose a and b

Solution:-

x	y	x · y	x ²
1	14	14	1
2	27	54	4
3	40	120	9
4	55	220	16
5	68	340	25
$\sum x = 15$	$\sum y = 204$	$\sum xy = 748$	$\sum x^2 = 55$

$$B = 13.6, A = 0$$

Curve fitting of a quadratic curve:

$$y = ax^2 + bx + c$$

$$\sum y = a \sum x^2 + b \sum x + nc \quad \text{(i)}$$

Multiply by 'x' on both sides;

$$\sum xy = a \sum x^3 + b \sum x^2 + c \sum x \quad \text{(ii)}$$

Multiply by 'x' on both sides;

$$\sum x^2 y = a \sum x^4 + b \sum x^3 + c \sum x^2 \quad \text{(iii)}$$

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Q:- Fit a second degree polynomial at 'y' on x to following data:

x	50	70	100	120
y	12	15	21	25

Solution:-

	x	y	x^2	x^3	x^4	xy	x^2y
	50	12	2500	125,000	6250000	600	30000
	70	15	4900	343,000	24010000	1050	73500
	100	21	10,000	100,000	100000000	2100	210000
	120	25	14,400	1728,000	207360000	3000	360000
Σ	340	73	31800	3196,000	337620000	6150	613500

$$B = 5.52586; A = 6.10293; C = 5 \times 10^{-7}$$

Curve fitting of a exponential curve;

$$y = ab^x$$

log on both sides;

$$\log y = \log ab^x$$

$$\log y = \log a + \log b^x$$

$$\log y = \log a + x \log b$$

$$\text{Let } A = \log a$$

$$B = \log b$$

$$Y = \log y$$

$$Y = A + Bx$$

Summation on both sides;

$$\Sigma Y = nA + B \Sigma x \quad \text{--- (i)}$$

Multiply by 'x' on both sides

$$\Sigma xy = A \Sigma x + B \Sigma x^2 \quad \text{--- (ii)}$$

* is more or less A, B, a, y, x, z, log, large to a b (small) y, x, z

Q. Find the exponential curve $y = ab^x$ from the given data.

x	2	4	6	8	10
y	1	3	6	12	24

$n = 5$

Solution:-

$\log_{10}()$

x	y	$y = \log y$	x^2	$x \cdot y$
2	1	0	4	0
4	3	0.4771	16	1.9084
6	6	0.7781	36	4.6686
8	12	1.0791	64	8.6328
10	34	1.3802	100	13.802
36	56	3.7145	220	29.0118

Substitute the value in eq (i) and eq (ii) to find A and B,

$$3.71 = 5A + 30B \quad \text{--- (i)}$$

$$29.013 = 30A + 220B \quad \text{--- (ii)}$$

On solving;

$$A = -0.266, B = 0.16815$$

For a, b;

$$A = \log a$$

$$B = \log b$$

$$\log^{-1}(-0.266) = a$$

$$\log^{-1}(0.16815) = b$$

$$a = 0.5420$$

$$b = 1.4788$$

Put in;

$$y = ab^x$$

$$y = (0.5420)(1.4788)^x$$

$$y = 0.1982$$

CHAPTER 6:- DIFFERENCE EQUATION

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General equation:-

$$a_0 y_{n+k} + a_1 y_{n+k-1} + a_2 y_{n+k-2} + \dots + a_k y_n = f(n)$$

where;

$a_0, a_1, a_2, \dots, a_k$ are coefficient

if $f(n) = 0 \rightarrow$ homogeneous

if $f(n) \neq 0 \rightarrow$ non homogeneous

Solving higher order difference equation (Homogeneous)
 $\Rightarrow$ 2nd order

$$a_0 y_{n+2} + a_1 y_{n+1} + a_2 y_n = 0$$

$$\text{Put } y_{n+2} = y^{n+2}$$

$$y_{n+1} = y^{n+1}$$

$$y_n = y^n$$

So,

$$a_0 y^{n+2} + a_1 y^{n+1} + a_2 y^n = 0$$

Take y^n common,

$$(a_0 y^2 + a_1 y + a_2) y^n = 0$$

$$a_0 y^2 + a_1 y + a_2 = 0$$

$\Rightarrow$ CHARACTERISTIC EQUATION

CASE 1:- Real and Distinct Roots: $r_1 \neq r_2$

$$y_h = A(r_1)^n + B(r_2)^n$$

CASE 2:- Real and Equal Roots: $r_1 = r_2$

$$y_h = (A + Bn)(r)^n$$

CASE 3:- Complex Roots: $r = \alpha \pm \beta i$

$$y_h = (r)^n (A \cos nx + B \sin nx)$$

where,

$$r = \sqrt{\alpha^2 + \beta^2}; \theta = \tan^{-1} \left(\frac{\beta}{\alpha} \right)$$

Q1- Solve $y_{n+2} - 5y_{n+1} + 6y_n = 0$
initially $y_0 = 1$ and $y_1 = 2$
Solution -

$$y^{n+2} - 5y^{n+1} + 6y^n = 0$$

$$y^n(y^2 - 5y + 6) = 0$$

$$y^2 - 5y + 6 = 0$$

$$y^2 - 2y - 3y + 6 = 0$$

$$y(y-2) - 3(y-2) = 0$$

$$(y-2)(y-3) = 0$$

$$y-2 = 0 ; y-3 = 0$$

$$y_1 = 2 \quad y_2 = 3$$

Here, $y_1 = 2$ and $y_2 = 3$

$y_n = A(y_1)^n + B(y_2)^n$ cuz roots are real and distinct.

$$y_n = A(2)^n + B(3)^n \quad \text{--- (1)}$$

Using initial condition $\Rightarrow y_0 = 1$

$$n=0 ; y_n = 1$$

$$1 = A + B \quad \text{--- (2)}$$

$$y_1 = 2$$

$$n=1 ; y_n = 2$$

$$2 = 2A + 3B \quad \text{--- (3)}$$

By solving, $A=1, B=0$

Put in eq (1),

$$y_n = (1)(2)^n + (0)(3)^n$$

$$y_n = 2^n$$

Q2- Solve $y_{n+2} - 4y_{n+1} + 4y_n = 0$
Solution -

$$y^{n+2} - 4y^{n+1} + 4y^n = 0$$

$$y^n(y^2 - 4y + 4) = 0$$

$$y^2 - 4y + 4 = 0$$

$$y^2 - 2y - 2y + 4 = 0$$

$$y(y-2) - 2(y-2) = 0$$

$$(y-2)(y-2) = 0$$

$$y-2 = 0 ; y-2 = 0$$

$$y_1 = 2 \quad y_2 = 2$$

Here, $y_1 = 2$ and $y_2 = 2$

$y_n = (A + Bn)(y)^n$ cuz roots are equal.

$$y_n = (A + Bn)2^n$$

Q3- Solve $y_{n+2} - 2y_{n+1} + 2y_n = 0$
Solution -

$$y^{n+2} - 2y^{n+1} + 2y^n = 0$$

$$y^n(y^2 - 2y + 2) = 0$$

$$y^2 - 2y + 2 = 0$$

$$y = 1 \pm i$$

$y_n = (A \cos n\theta + B \sin n\theta)$ cuz roots are complex.

$$\text{Here, } y = \sqrt{a^2 + b^2} = \sqrt{(1)^2 + (1)^2} = \sqrt{2}$$

$$\theta = \tan^{-1} \frac{b}{a} = \tan^{-1} \left(\frac{1}{1} \right) = \frac{\pi}{4}$$

$$y_n = (\sqrt{2})^n \left(A \cos \frac{n\pi}{4} + B \sin \frac{n\pi}{4} \right)$$